Task 1: Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

The OSI model (Open Systems Interconnection model) is a conceptual framework that describes how data travels from one device to another over a network.

Conceptual means it's a model for *understanding*, not a physical thing you can touch or a piece of software you install.

What OSI is NOT:

- It's not a protocol itself (you can't "run OSI" on a computer)
- It's not a physical set of components
- No real network strictly implements all 7 layers exactly as below

Layer 7	Application	HTTP, HTTPS, FTP, SMTP — e.g. Typing a website in web browser
Layer 6	Presentation	SSL/TLS encryption, JPEG, ASCII — its encryption of the website, the lock icon 🔒 . So data is coded and unreadable to outsiders during transfer.
Layer 5	Session	NetBIOS, RPC — e.g. a login session between apps. This layer manages starting, maintaining, and ending that connection.
Layer 4	Transport	TCP, UDP — e.g. When you download a file and it arrives complete and error-free (not corrupted or missing parts) — that's because this layer (TCP) breaks the file into small chunks, numbers them, and makes sure every chunk arrives and gets reassembled correctly.
Layer 3	Network	IP, ICMP — e.g. a Router. The ISP decides the best route for the data from PC to server. Routing is based on IP address
Layer 2	Data Link	MAC addresses, Ethernet — e.g. a Switch. When your laptop connects to a Wi-Fi **router** or LAN and it recognizes your device specifically among all the devices in the network (using your device's unique MAC address) — that's this layer at work, handling local delivery.
Layer 1	Physical	Cables, Hubs, NICs — e.g. an Ethernet cable. physical medium data travels through.

Task 2 : Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames. Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

a) Encrypting data before sending: **Presentation (Layer 6)**
b) Routing packets across networks: **Network (Layer 3)** routing decisions (finding the best path across networks using IP addresses) happen at the Network layer
c) Converting data to electrical signals: **Physical (Layer 1)**
d) Opening a web page in your browser: **Application (Layer 7)**
e) Detecting and correcting errors in frames: **Data Link (Layer 2)** error-checking methods like CRC (Cyclic Redundancy Check) to detect and sometimes correct transmission errors at the local network level.

Task 3: Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

When you send a WhatsApp message:

1. You type it (**Application**)
2. It gets encrypted (**Presentation**)
3. Your chat session with the WhatsApp server stays active (**Session**)
4. The message is broken into packets and tracked for delivery (**Transport**)
5. Those packets are routed across the internet to reach WhatsApp's servers (**Network**)
6. Your Wi-Fi router sends it locally over your home network first (**Data Link**)
7. It physically travels as radio waves from your phone to your router (**Physical**)

Task 4: Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Ping uses a protocol called ICMP (Internet Control Message Protocol), which operates at the Network layer (Layer 3)

Ping is a **diagnostic tool**, not an application like a browser. It doesn't need reliability, ports, sessions, or encryption. So it **skips Transport, Session, and Presentation entirely** and goes almost straight from the ping command to the Network layer.

Since ping to the IP address is successful, it confirms these layers are working correctly:

- Physical (Layer 1) — the cable/Wi-Fi signal is reaching the destination
- Data Link (Layer 2) — local network delivery (MAC addressing) is working
- Network (Layer 3) — IP addressing and routing are working, since ICMP (Network layer) successfully reached the server and got a reply.

Since the website itself fails to load, the problem must lie in one of the layers above Network:

- Transport (Layer 4) — could be a blocked or filtered port (e.g., port 80/443 blocked by a firewall), preventing a TCP connection from establishing even though basic connectivity works
- Session (Layer 5) — a session might fail to establish between browser and server
- Presentation (Layer 6) — SSL/TLS certificate issues could prevent the connection (common cause — e.g., an expired or misconfigured HTTPS certificate)
- Application (Layer 7) — the most likely everyday cause: DNS resolution issues. The web server itself might be down, or there's a misconfiguration in HTTP/HTTPS handling